

ECON 108

PRINCIPLES OF ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

CONSUMER CHOICE

CHAPTER 10 (from the book)

CHAPTER OUTLINE

Utility Theory

- Total Utility
- Marginal Utility
- Diminishing Marginal Utility

Consumer Choice

- Budget line
- Consumer Optimum

CHAPTER OUTLINE

Changes in Income and Consumer Optimum

Changes in Prices and Consumer Optimum

- Income effect
- Substitution effect

UTILITY THEORY

Utility Theory

- The analysis of consumer decision making based on utility maximization

Utility

- The want-satisfying power of a good or service. Utility is measured in “utils.”
 - Utility is at best an ordinal concept although here we will treat it as a cardinal concept.
 - We cannot compare utility across individuals.

UTILITY THEORY

Marginal Utility

- The change in total utility due to a one-unit change in the quantity of a good or service consumed

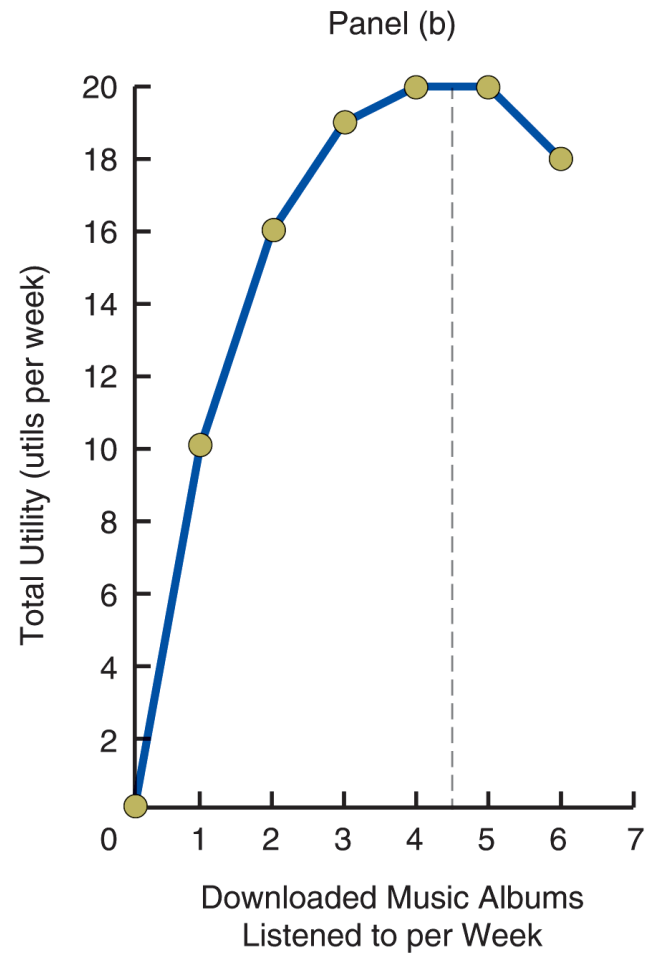
$$\text{Marginal utility} = \frac{\text{Change in total utility}}{\text{Change in number of units consumed}}$$

UTILITY THEORY

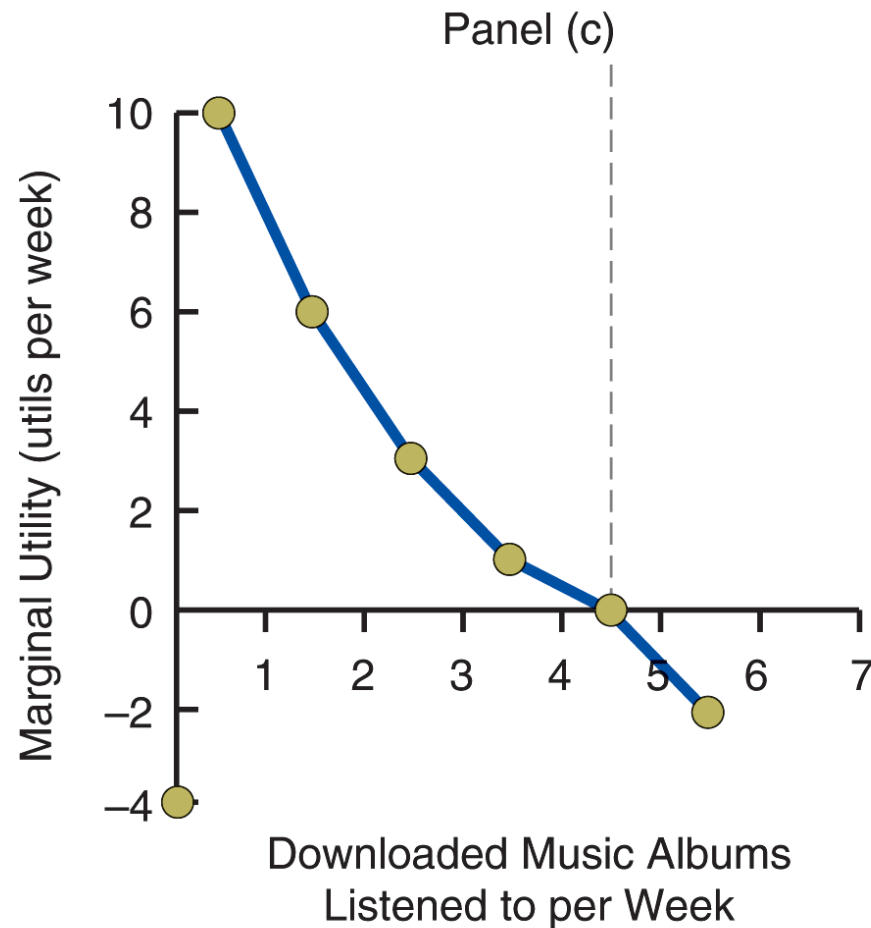
Panel (a)

(1) Number of Music Albums Downloaded and Listened to per Week	(2) Total Utility (utils per week)	(3) Marginal Utility (utils per week)
0	0	
1	10	10 (10 - 0)
2	16	6 (16 - 10)
3	19	3 (19 - 16)
4	20	1 (20 - 19)
5	20	0 (20 - 20)
6	18	-2 (18 - 20)

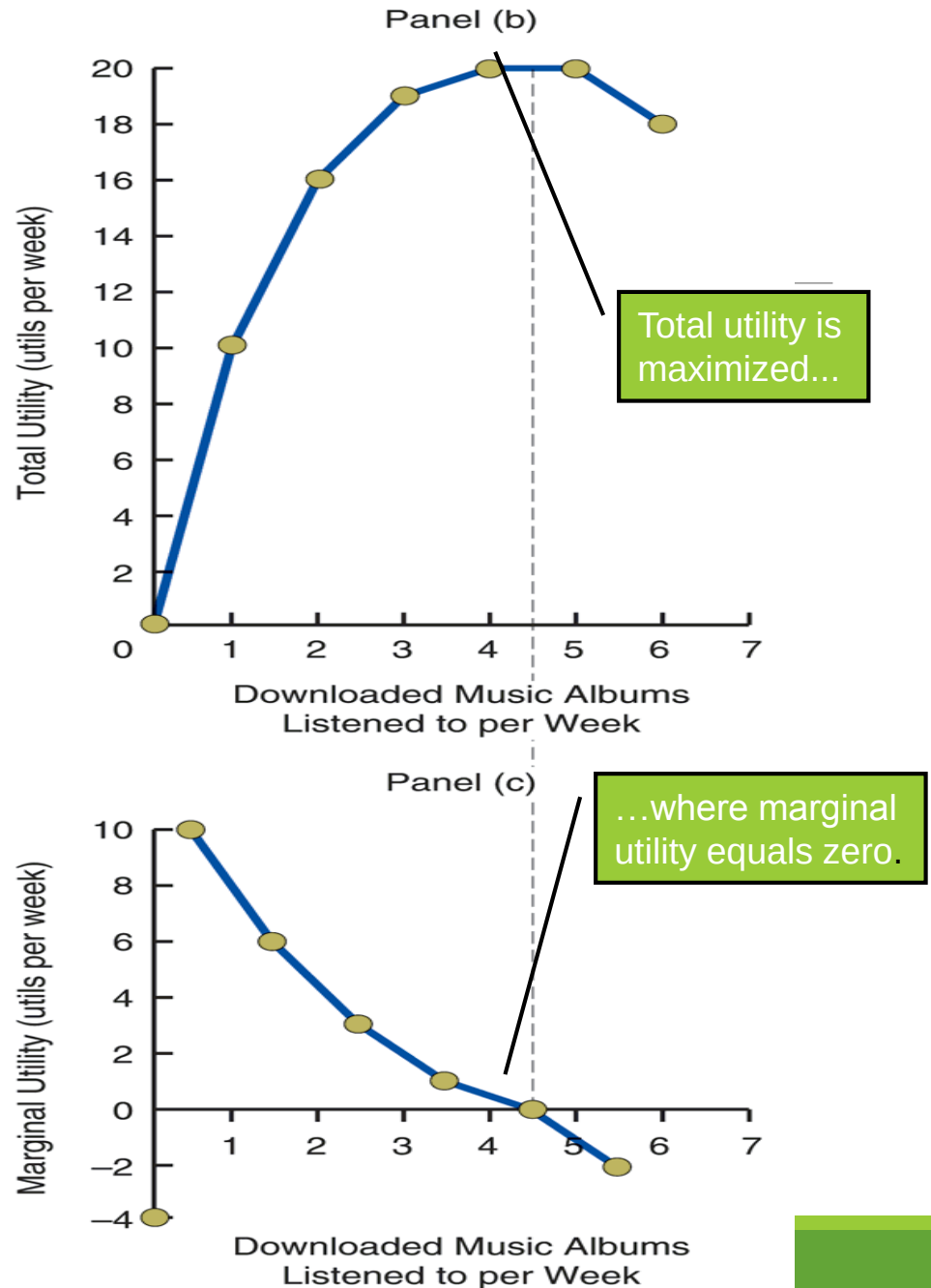
UTILITY THEORY



UTILITY THEORY



Total and Marginal Utility of Downloading and Listening to Digital Music Albums.



UTILITY THEORY

Observations

- Marginal utility falls as more is consumed
- Marginal utility is zero when total utility is maximized (satiation point)
- Marginal utility is negative when total utility falls

DECREASING MARGINAL UTILITY

Decreasing marginal utility

- The principle that as more of any good or service is consumed, its extra benefit declines
- Increases in total utility from consumption of a good or service become smaller and smaller as more is consumed during a given time period.

Example: Newspaper Vending Machines versus Candy Vending Machines

How many people take more than one paper from the vending machine?

Why not dispense candy the same way?

The answer is found in the concept of diminishing marginal utility.

CONSUMER CHOICE

Basic setup

- We will analyze the problem of a consumer trying to allocate a fixed income between two commodities.
- The parameters in the problem (the given) are: income, price of good one, and price of good two.
- The choice variables in the problem (the unknown) are: quantity of good one, and quantity of good two.

CONSUMER CHOICE

The budget line

- Monetary income : M
- Price of good 1: P_1
- Price of good 2: P_2

CONSUMER CHOICE

Consumer Optimum

- Is the point along the budget line for which the total utility is maximized.
- With only one good utility is maximized by spending all income in that good (as long as marginal utility is positive)
- With two goods we need to weight the marginal utility of either good.

CONSUMER CHOICE

Marginal Utility

- Is the change in total utility due to a one-unit change in the **quantity of a good consumed**

Weighted Marginal Utility

- Is the change in total utility due to a one-unit change in the **amount of money spent on the good**

CONSUMER CHOICE

(1) Music Album Downloads per Period	(2) Total Utility of Music Album Downloads per Period (utils)	(3) Marginal Utility (utils) MU_d	(4) Marginal Utility per Dollar Spent (MU_d/P_d) (price = \$5)
0	0	—	—
1	50.0	50.0	10.0
2	95.0	45.0	9.0
3	135.0	40.0	8.0
4	171.5	36.5	7.3
5	200.0	28.5	5.7

CONSUMER CHOICE

(5)	(6)	(7)	(8)
Sandwiches per Period	Total Utility of Sandwiches per Period (utils)	Marginal Utility (utils) MU_s	Marginal Utility per Dollar Spent (MU_s/P_s) (price = \$3)
0	0	—	—
1	25	25	8.3
2	47	22	7.3
3	65	18	6.0
4	80	15	5.0
5	89	9	3.0

CONSUMER CHOICE

(1)	(4)	(8)
Music Album Downloads per Period	Marginal Utility per Dollar Spent (MU_d/P_d) (price = \$5)	Marginal Utility per Dollar Spent (MU_s/P_s) (price = \$3)
0	—	—
1	10.0	8.3
2	9.0	7.3
3	8.0	6.0
4	7.3	5.0
5	5.7	3.0

CONSUMER CHOICE

Optimizing Condition

- A consumer's money income should be allocated so that the last dollar spent on each good purchased yields the same amount of marginal utility (when all income is spent)
- This rule yields the largest possible total utility.

CONSUMER CHOICE

- Mathematically

- All income should be spent:

$$M = P_1X_1 + P_2X_2$$

- The marginal utility per dollar spent on each good should be the same:

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$

CONSUMER CHOICE

Choices

Purchase	Music Album Downloads		Sandwiches	
	Unit	MU_d/P_d	Unit	MU_s/P_s
1	First	10.0	First	8.3
2	Second	9.0	First	8.3
3	Third	8.0	First	8.3
4	Third	8.0	Second	7.3
5	Fourth	7.3	Second	7.3

CONSUMER CHOICE

Purchase	Buying Decision	Remaining Income
1	First music album download	$\$26 - \$5 = \$21$
2	Second music album download	$\$21 - \$5 = \$16$
3	First sandwich	$\$16 - \$3 = \$13$
4	Third music album download	$\$13 - \$5 = \$8$
5	Fourth music album download and Second sandwich	$\$8 - \$5 = \$3$ $\$3 - \$3 = \$0$

CHANGES IN INCOME

Changes in income and consumer optimum

- An increase (a decrease) in income shifts the budget line to the right (left) allowing the consumer to consume more (less) and thus increasing (decreasing) consumer's utility.

CHANGES IN PRICE

Changes in price and consumer optimum

- A decrease (an increase) in price rotates the budget line to the right (left) allowing the consumer to consume more (less) and thus increasing (decreasing) consumer's utility.

CHANGES IN PRICE

Recall from the table above Income = \$26

$$Q_d = 4 \quad \longrightarrow \quad \frac{MU_d}{P_d} = \frac{36.5}{5} = 7.3$$

$$Q_s = 2 \quad \longrightarrow \quad \frac{MU_s}{P_s} = \frac{22}{3} = 7.3$$

CHANGES IN PRICE

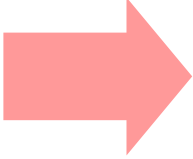
Assume Price of Music Falls to \$4

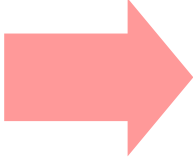
$$Q_d = 4 \longrightarrow \frac{MU_d}{P_d} = \frac{36.5}{4} = 9.125$$

$$Q_s = 2 \longrightarrow \frac{MU_s}{P_s} = \frac{22}{3} = 7.3$$

CHANGES IN PRICE

Assume Price of Music Falls to \$4

Now  $\frac{MU_d}{P_d} > \frac{MU_s}{P_s}$

Result  Buy more downloads
and MU_d falls

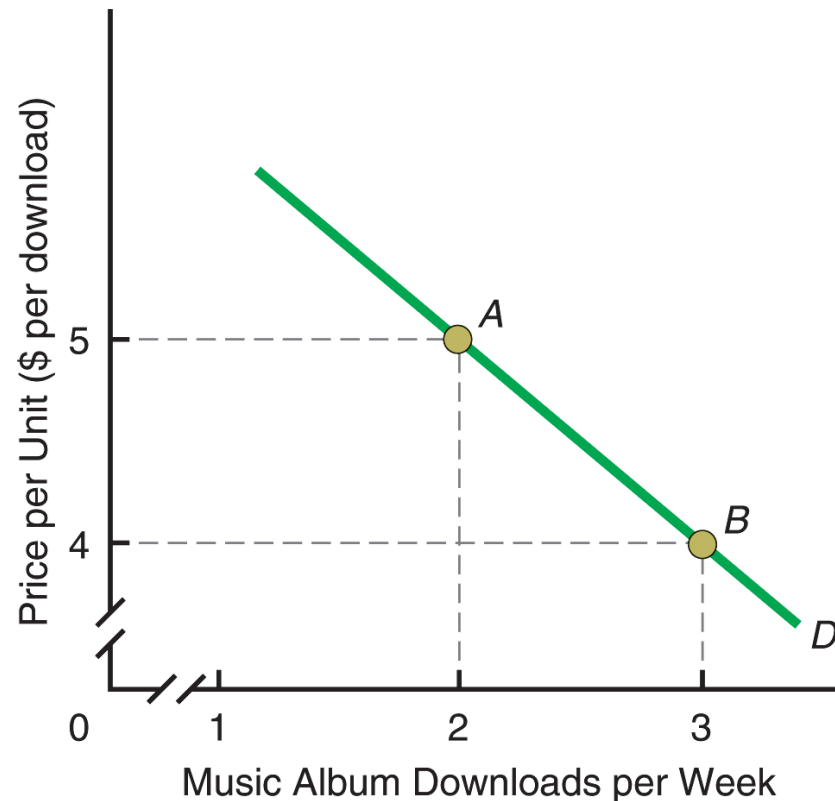
CHANGES IN PRICE

Changes in price and consumer demand

- The consumer choice is consistent with the law of demand: *amount purchased is inversely related to price.*

CHANGES IN PRICE

Changes in price and consumer demand



CHANGES IN PRICEb

Income and Substitution effects of a price change

- Every price change induces changes in the relative price (i.e., ratio of prices), and in the real income (i.e., the purchasing power of the monetary income)

CHANGES IN PRICE

Substitution effect

- **The substitution effect** occurs when a consumer buys more of a good as a result of a relative price change.
- Consumers and producers shift away from goods and resources that become priced relatively higher in favor of goods and resources that are now priced relatively lower.

CHANGES IN PRICE

Real-income effect

- The **real-income effect** occurs when there is a change in purchasing power as a result of a change in the price of a good.
- When the price goes up (down), real income, or purchasing power, falls (increases).

END OF THE CHAPTER